

Arpit Sharma

CONTACT INFORMATION

 Suisun City, CA 94585
 [Google Scholar Profile](#)
 arpit7123@gmail.com

 (480)466-3889
 [arpitsharma2](#)

SUMMARY

Computer Scientist with 6+ years' experience designing and deploying scalable conversational AI and agentic systems. Proven innovator with patented contributions. Strategic leader skilled in mentoring teams and shaping corporate strategy to deliver AI-driven solutions for major retail use cases. Expertise in Large Language Models (LLMs), agentic frameworks, and AI workflow orchestration.

RELEVANT WORK EXPERIENCE

Walmart Associates Inc., Sunnyvale, CA, USA

Distinguished Data Scientist, June 2025 - Present

- Led the Intelligence team in the successful launch of multiple features within Walmart's Shopping Assistant, as well as several Minimum Viable Products (MVPs).
- Designed and implemented multi-agent workflows to support innovative shopping use cases, including event planning and essentials shopping, enhancing user experience and automation.
- Spearheading industry collaboration research initiatives focused on the development of low-cost, high-quality open-source LLM (Large Language Model) judges.

Principal Data Scientist, October 2022 - June 2025

- Provided technical leadership to a cross-functional team of over 15 engineers and scientists, guiding the multi-phase delivery of advanced machine learning systems.
- Designed and implemented an agentic evaluation framework for internal assessment of agentic workflows within a retail-focused chatbot.
- Defined quality and performance metrics for large language models (LLMs) across various retail use cases; additionally, developed an "LLM-as-a-judge" framework to evaluate these metrics.
- Led the fine-tuning of open-source LLMs to meet business-specific requirements for retrieval-augmented generation (RAG) and code-of-conduct compliance, utilizing techniques such as supervised fine-tuning (SFT) and direct preference optimization (DPO).
- Drove cross-organizational alignment by coordinating priorities, timelines, and success criteria across product, engineering, science, and operations teams—effectively resolving conflicting objectives and ensuring the timely delivery of high-impact solutions.
- Defined a schema for AI agents in retail environments and used it to develop multiple agent types, including RAG, Instruct, and Planning agents.
- Successfully led an academic research collaboration that resulted in a paper accepted at the international NLP conference NAACL 2025.

Staff Data Scientist, December 2020 - October 2022

- Developed and fine-tuned a BERT-based model for Natural Language Understanding (NLU) in a conversational AI application.
- Led the design and implementation of a clustering-based, self-serve intent prediction model for multi-domain chatbots, leveraging the Universal Sentence Encoder (USE) for text embeddings.
- Spearheaded a research initiative focused on domain-specific adaptation of Distil-BERT for retail applications, incorporating universal dependency parsing techniques.

- Designed API contracts for the core intelligence module powering Walmart's conversational AI platform.

Senior Data Scientist, September 2019 - December 2020

- Trained a Bi-LSTM + CRF model for intent classification and named entity recognition in Walmart's shopping assistant.
- Researched and developed a deep learning-based intent classification model with contextual awareness for improved performance in multi-turn conversations.
- Designed and implemented a text normalization pipeline to preprocess and clean ill-formatted user inputs for Walmart's shopping assistant, enhancing downstream model accuracy.

Data Scientist Intern, May 2018 - August 2018

- Enhanced eCommerce efficiency by developing semantic parsers for customer queries.
- Improved query response relevance through detailed analysis, boosting user satisfaction.
- Collaborated in team meetings, fostering innovation and knowledge sharing.

Mathworks, Natick, MA, USA

Application Support Engineer Intern, May 2013 - Aug 2013

- Provided technical support for MATLAB toolboxes. Dealt with technical issues involving parallel processing, image processing, MATLAB external interfaces and MEX files.

Infosys, Mysore, INDIA

Systems Engineer, Sep 2010 - Apr 2012

- Developed and supported financial system for insurance agents. Worked on Java, PL/SQL, UNIX, COBOL and Mainframe.

TECHNICAL SKILLS

Agentic AI, Conversational Systems & Machine Learning: Multi-Agent Systems, Google ADK, RAG, LLM fine-tuning, Transformers, prompt engineering, RLHF, RLAIF, PPO, DPO and GRPO for LLM alignment. Full ML life-cycle management. Expertise in advanced ML algorithms and deep learning frameworks (PyTorch, TensorFlow).
Core Technology & Architecture: Designing and building low-latency, fault-tolerant AI systems. Proficiency with GCP and Docker.
Ethical AI & Evaluation: Implementing safety, privacy, and compliance measures for conversational agents. Developing robust frameworks for evaluating and debugging complex agent behavior.
Programming: Java, Python, MATLAB, HTML, MongoDB, Oracle SQL, Neo4j.

EDUCATION

Arizona State University, Tempe, AZ, USA

Ph.D., Computer Science, January 2015 - August 2019

- Thesis Topic: *Towards Understanding Natural Language: Semantic Parsing, Commonsense Knowledge Acquisition and Applications*

M.S., Computer Science, August 2012 - December 2014, *Suma Cum Laude*

- Thesis Topic: *Solving Winograd Schema Challenge : Using Semantic Parsing, Automatic Knowledge Acquisition and Logical Reasoning*

College Of Engineering Roorkee, Roorkee, INDIA

B.Tech., Computer Science, August 2006 - May 2010

- Grade: *75.2 on a scale of 100*
- *B.Tech with honors*